

DiaFoot.AI — Complete Project Report

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title: “DiaFoot.AI — Complete Project Report” subtitle: “Diabetic Foot Ulcer Detection, Segmentation & Wagner Staging (v2.0.0)” author: “Ru

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Package: `diafootai` v2.0.0 · Python 3.12–3.13 · PyTorch 2.10 **License:** MIT · **Status:** Beta (research code, not a cleared medical device) **Report date:** 2026-07-17

Data-provenance and honesty note (read first). Every performance number in this report comes from the machine-readable evaluation artifacts in `results/*.json` and `data/metadata/*.json`, computed on **leakage-audited clean splits**. These numbers supersede the headline in `CHANGELOG.md` (v2.0.0, Dice 85.89%), which is a DFU-only, single-skin-tone subgroup figure produced **before** the data-leakage fix. Where an analysis was run on a different split or an earlier checkpoint, that provenance is stated inline. Nothing here is rounded up for effect.

1. Executive summary

DiaFoot.AI classifies a foot photograph as `Healthy`, `Non-DFU`, or `DFU`, segments the wound when one is present, and measures wound area in mm². It is a **cascaded multi-task** pipeline built to be honest about its limits.

Headline results (leakage-audited test set, n = 1,161):

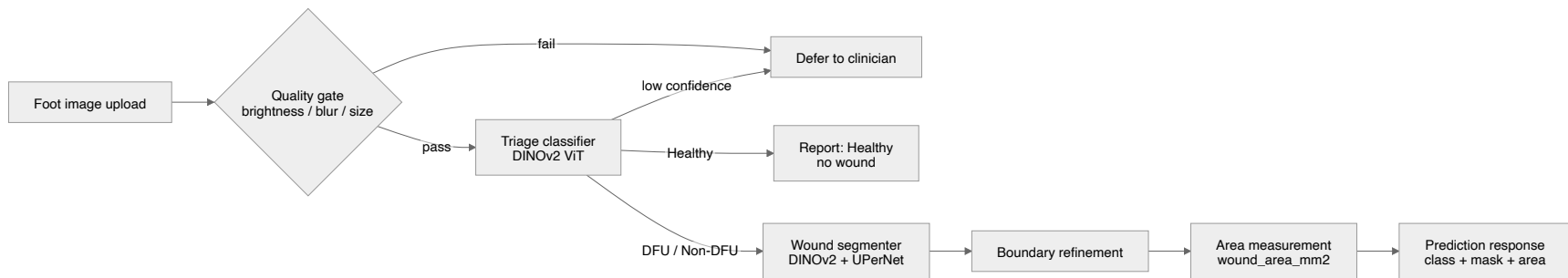
Capability	Metric	Value
Triage classification	Accuracy	98.4%
Triage classification	DFU sensitivity	96.6%
Triage classification	Macro AUROC	0.999
Wound segmentation (DFU wounds)	Dice	0.891
Wound segmentation (5-fold CV)	Dice	0.853 ± 0.009
Boundary quality (DFU)	HD95	11.3 px
Calibration	ECE (after temp. scaling)	0.007
Deployment	ONNX speedup	4.5x

Two caveats define the project’s integrity: (1) on the **full mixed** test set the segmentation *mean* Dice falls to 0.65–0.72 because empty-mask images punish any false positive (median stays 0.93); (2) the triage **classifier does not generalize** to unseen image sources (external accuracy 21%, DFU sensitivity 0%). Both are documented in full below.

2. Problem and clinical motivation

Diabetic foot ulcers are a leading cause of lower-limb amputation. Automated triage plus wound measurement can support earlier intervention. DiaFoot.AI **v1** segmented every image as if it contained an ulcer — it had no concept of “not a wound”, so it hallucinated ulcers on healthy skin. **v2** puts a classifier in front so the system can abstain on healthy skin, and adds a defer-to-clinician gate so low-confidence or low-quality inputs go to a human instead of producing an unreliable answer.

PIPELINE OVERVIEW



3. Dataset

3.1 COMPOSITION (VERIFIED IMAGE/MASK COUNTS)

Category	Images	Masks	Paired	Mask type
DFU (incl. AZH)	2,119	2,119	2,119	Wound boundary
Healthy feet	3,300	3,300	3,300	Empty (all-zero)
Non-DFU conditions	2,686	2,686	2,686	Non-DFU wound boundary
Total	8,105	8,105	8,105	—

All images are 512×512 after preprocessing (0 orphans, 0 non-512 in sample). The AZH wound-care dataset (1,109 images) is integrated into the DFU category. Sources span FUSeg, AZH, Kaggle, and Mendeley (healthy/non-DFU).

3.2 SKIN-TONE DISTRIBUTION (ITA — INDIVIDUAL TYPOLOGY ANGLE)

Computed for fairness auditing. Higher ITA = lighter skin.

Dataset	Analyzed	Mean ITA	Dominant categories
DFU (FUSeg subset)	810	−50.55	Dark 633, Brown 72, Very Light 56
Healthy	3,300	62.00	Very Light 2,472, Brown 220, Dark 180
Non-DFU	2,686	−7.02	Dark 931, Brown 685, Tan 401

The DFU images skew toward darker skin tones; healthy images skew light. This imbalance is why fairness is audited on the DFU-only slice (Section 5.6).

3.3 SPLITS AND LEAKAGE AUDIT

Final splits are **70/15/15, doubly stratified** by ITA skin tone and class:

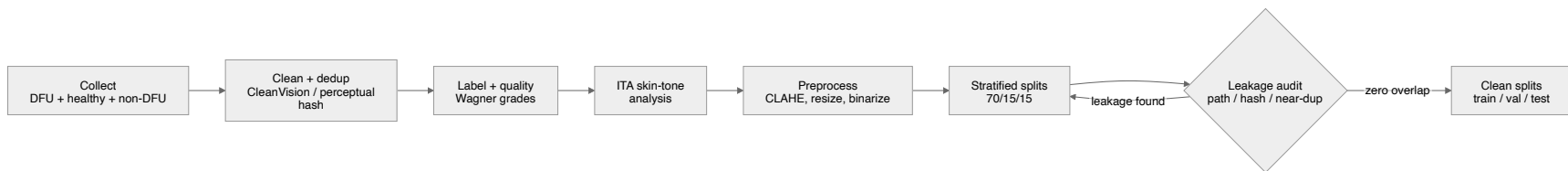
Split	Count	Classes
Train	5,782	Healthy + Non-DFU + DFU
Val	1,162	—
Test	1,161	Healthy 495 · Non-DFU 403 · DFU 263

Leakage fix (the integrity story). A perceptual-hash audit of the pre-fix splits found large numbers of near-duplicate image pairs straddling train and test — a model tested partly on images it effectively trained on reports inflated scores. The splits were rebuilt from scratch and re-audited.

Overlap axis	Before (baseline)	After (clean recheck)
Path overlap (trainxtest)	0	0
Canonical-ID overlap (trainxtest)	0	0
Content-hash overlap (trainxtest)	0	0
Near-duplicate pairs (trainxtest)	96,829	0
Near-duplicate pairs (trainxval)	13,557	0
Near-duplicate pairs (valxtest)	2,375	0
<code>has_any_leakage</code>	True	False

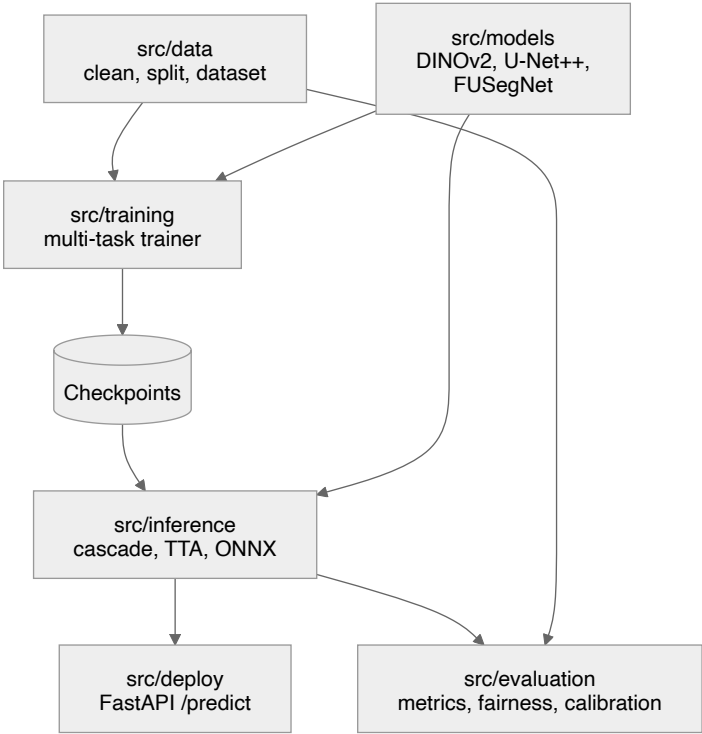
All results in Sections 4–5 are on the clean (`has_any_leakage: False`) splits.

3.4 DATA PIPELINE



4. Method and architecture

Two model families are implemented. The **deployed** path is DINOv2-based; U-Net++ / FUSegNet / nnU-Net exist for research and ablation.



Stage	Deployed model	Research alternatives
Triage classifier	DINOv2 ViT + 3-class head (LoRA rank 8, α 16)	EfficientNet-V2-M
Wound segmenter	DINOv2 + UPerNet decoder	U-Net++ (EfficientNet-B4, scSE), FUSegNet (EfficientNet-B7, P-scSE), nnU-Net v2
Post-processing	Morphological + connected-component boundary refinement	—

TRAINING CONFIGURATION (ACTUAL VALUES)

Setting	Value
Seed	42
Image size	512×512 (deploy: 518×518 for DINOv2)
Batch size	16 (baseline) / 24 (multi-task)
Optimizer	AdamW, lr 1e-4, weight decay 1e-2
Scheduler	Cosine annealing, 10-epoch linear warmup
Precision	BFloat16 mixed (H200)
Segmentation loss	DiceCE / Dice-boundary compound
Classification loss	Focal
Staging loss	Label-smoothing CE (0.1)
EMA decay	0.999
Early stopping	patience 15
Multi-task weights	classification 1.0 · segmentation 2.0 · staging 1.0 (GradNorm dynamic)
Curriculum	classifier warmup 10 ep · seg unfreeze ep 10 · staging unfreeze ep 20

5. Results (actual values)

5.1 TRIAGE CLASSIFICATION — TEST SET (N = 1,161)

Metric	Value
Accuracy	0.9836
Macro F1	0.9812
Weighted F1	0.9836
Macro precision	0.9820
Macro recall	0.9805
Macro AUROC	0.9991
DFU sensitivity (recall)	0.9658
Healthy specificity	0.9955

Per class:

Class	Precision	Recall	F1	AUROC	Support
Healthy	0.9940	0.9980	0.9960	1.0000	495
Non-DFU	0.9752	0.9777	0.9765	0.9984	403
DFU	0.9769	0.9658	0.9713	0.9989	263

Calibration and abstention:

Metric	Value
ECE before temperature scaling	0.0394
ECE after (T = 0.4)	0.0075
Brier before → after	0.0344 → 0.0258
Defer threshold	0.95
Coverage at threshold	0.9345 (defer rate 0.0655)
Kept / deferred cases	1,085 / 76
Accuracy on kept cases	0.9972

The defer gate trades 6.5% coverage for 99.7% accuracy on the cases it keeps.

5.2 WOUND SEGMENTATION — TEST SET

Slice	Dice	IoU	HD95 (px)	NSD@2mm	NSD@5mm
DFU wounds only (n = 263)	0.891	0.829	11.3	0.888	0.972
Full mixed set (n = 1,161), mean	0.718	0.673	66.1	0.643	0.713
Full mixed set , median	0.929	0.868	5.0	0.887	1.000
5-fold CV (DFU)	0.853 ± 0.009	0.785 ± 0.010	—	—	—

Why mean « median on the mixed set: healthy and non-DFU images have empty ground-truth masks; Dice on an empty mask is 0 if the model predicts even one false-positive pixel. A handful of such near-zero scores collapse the mean while the median (dominated by real wounds) stays high. Judge wound-outlining quality from the DFU-only mean (0.891) and the median (0.929).

5.3 ARCHITECTURE / DATA-COMPOSITION COMPARISON (DFU DICE)

Model / data	DFU Dice	DFU IoU	NSD@5mm
U-Net++ (DFU-only)	0.851	0.775	0.952
U-Net++ (All classes)	0.824	0.737	0.933
FUSegNet (DFU + non-DFU)	0.818	0.730	0.923
U-Net++ v2 (DFU + non-DFU, fixed)	0.804	0.707	0.908
U-Net++ (DFU + non-DFU)	0.790	0.690	0.912

Training on DFU-only gives the highest DFU Dice, but that model cannot triage; the deployed cascade accepts a small DFU-Dice cost for the ability to reject healthy/non-DFU inputs.

5.4 CLINICAL WOUND-AREA AGREEMENT (N = 3, INDICATIVE ONLY)

Metric	Value
MAE	7.23 mm ²
RMSE	9.86 mm ²
MAPE	3.28%
Bias	+5.70 mm ²
Pearson r	0.9997

The near-perfect correlation is on **only 3** manually measured wounds — treat as indicative, not validated.

5.5 TEST-TIME AUGMENTATION (200-IMAGE SUBSET)

Config	Dice	IoU	HD95
Base	0.574	0.523	87.5
TTA (16 augmentations)	0.613	0.563	84.9

TTA adds +3.9% Dice. The absolute values are low because this subset is mixed (empty-mask images included).

5.6 FAIRNESS (ITA SKIN TONE)

Slice	Group	n	Dice	Gap	Bias concern
DFU-only	Brown	285	0.859	0.00	No
All images	Brown	915	0.582	0.114	Flagged
All images	Tan	1	0.468	—	—

Honest reading: the DFU-only fairness gap is 0.00. The all-images “gap” of 0.114 is the difference between the Brown group (n = 915) and the Tan group (**n = 1**) — it is driven by a single image and is not statistically meaningful. The test set is effectively one ITA group (`brown / unknown`), so a robust cross-tone fairness claim cannot be made from this split; broader skin-tone evaluation is future work.

5.7 EXTERNAL VALIDATION (UNSEEN DATASETS)

Task	Internal	External	Absolute drop
Classification accuracy	1.000	0.210	−0.790
Classification macro-F1	1.000	0.333	−0.667
DFU sensitivity	1.000	0.000	−1.000
Healthy specificity	1.000	0.989	−0.011
Segmentation Dice (DFU)	0.891	0.893	+0.002 (no drop)
Segmentation HD95 (DFU)	11.3	9.5	improves

The single most important limitation: the **segmenter transfers** to new wound datasets (external DFU Dice 0.893, n = 552, essentially no drop), but the **triage classifier collapses** on out-of-distribution sources — external accuracy 21%, DFU sensitivity 0%, external per-class F1 Healthy 0.965 / Non-DFU 0.035 / DFU 0.000. Do not deploy the classifier on a new image source without

re-validating on that source. (The internal accuracy shown here is from the external- validation run’s own checkpoint; the calibrated production classifier scores 0.984 internally, Section 5.1.)

5.8 SEGMENTATION FAILURE ATLAS (DFU TEST SUBSET, N = 319)

Category	Count
Acceptable	302
Boundary error	7
False positive on empty GT	8
Poor overlap	2

95% of DFU test segmentations are acceptable; the dominant failure mode is false positives on empty ground truth (8) — consistent with the mixed-set mean-vs-median gap in Section 5.2. (Run on an earlier DFU-only test split of 319 images.)

5.9 DEPLOYMENT PERFORMANCE (ONNX PARITY)

Metric	Value
Samples	128
Mask agreement (ONNX vs PyTorch)	99.99% (min 99.996%)
Mean absolute output diff	0.0012
PyTorch latency	1,055.8 ms
ONNX latency	235.0 ms
Speedup	4.5x

5.10 SHORTCUT-RELIANCE AUDIT (SUPPLEMENTARY, PROVENANCE CAVEAT)

Run on an **earlier classifier checkpoint** (`best_epoch004_1.0000.pt` , 100% baseline accuracy) over 319 images, so read the *relative* robustness, not the absolute accuracy:

Perturbation	Accuracy drop	Prediction consistency
Border noise	3.4%	96.6%
Center-only (background removed)	0.0%	100%
Blur background	0.3%	99.7%

The near-zero drop when the background is removed or blurred suggests the classifier keys on the foot/wound region rather than background shortcuts. Because the checkpoint predates the leakage- clean classifier, treat this as a directional signal only.

6. Deployment

`uvicorn src.deploy.app:app --host 0.0.0.0 --port 8000` exposes:

Method	Path	Description
GET	<code>/health</code>	Liveness + whether the model is loaded
GET	<code>/model/info</code>	Model metadata, thresholds, limits
POST	<code>/predict</code>	Upload image → classification, wound stats, area, defer flags

`/predict` enforces upload guards (content type, max size), an image-quality gate (brightness/blur/size → `defer_to_clinician`), and a configurable rate limit. The response carries `classification`, `classification_confidence`, `classification_probs`, `defer_to_clinician`, `defer_reason`, `quality_flags`, `has_wound`, `wound_area_mm2`, `wound_coverage_pct`, and `inference_time_ms`. Deploy via uvicorn, Docker Compose (`docker/`), or the exported ONNX model.

7. Limitations

1. **Not a medical device.** No regulatory clearance; not for diagnosis or treatment.
2. **Triage classifier does not generalize across image sources** (external accuracy ~21%, DFU sensitivity 0%). Re-validate before any new-source use.
3. **Segmentation mean vs median gap:** aggregate mean Dice on mixed data is depressed by empty-mask false positives; judge wound performance from DFU-only / median numbers.
4. **Fairness is under-powered by the test split:** the mixed-set gap (0.114) is driven by a single `tan` image; DFU-only gap is 0.00. Broad skin-tone fairness is unproven.
5. **Small clinical-agreement sample** (area agreement $n = 3$).
6. **Some components are implemented but untrained** (MedSAM2 LoRA, nnU-Net v2).
7. **Mixed provenance in supplementary analyses** (shortcut audit / failure atlas use an earlier split or checkpoint), flagged inline.

8. Reproducibility

Item	Value
Seed	42
PyTorch	2.10.0
NumPy / SciPy / scikit-learn	1.26.4 / 1.15.1 / 1.6.1
OpenCV	4.11.0
Repro bundle generated	2026-03-11

Reproduce the honest re-run end to end:

```
python scripts/run_data_pipeline.py          # clean → ITA → splits
python scripts/run_leakage_audit.py          # verify has_any_leakage == false
python scripts/train.py --config configs/training/dinov2_baseline.yaml
python scripts/evaluate.py --task classify --checkpoint <ckpt>
python scripts/evaluate.py --task segment --checkpoint <ckpt>
```

See `docs/HPC_HONEST_RERUN_RUNBOOK.md` for the HPC procedure.

9. Repository and documentation index

```
DiaFoot.AI/
├─ src/{data,models,training,evaluation,inference,deploy}/
├─ scripts/          34 CLI entry points
├─ configs/          data / model / training / ablation / deploy YAML
├─ slurm/            HPC batch jobs
├─ tests/            pytest suite (244 passing, -m "not gpu")
├─ results/          verified evaluation JSON (source of every number here)
└─ docs/             Diataxis documentation
```

Document	Purpose
<code>README.md</code>	Overview + honest results
<code>docs/tutorial-getting-started.md</code>	Install → first prediction
<code>docs/howto-run-data-pipeline.md</code>	Build leak-free splits
<code>docs/howto-train.md</code>	Train and evaluate
<code>docs/howto-serve-api.md</code>	Serve / deploy the API
<code>docs/reference-cli.md</code>	Every CLI script + flags
<code>docs/reference-api.md</code>	REST endpoints and schemas
<code>docs/reference-architecture.md</code>	<code>src/</code> module map
<code>docs/explanation-pipeline-design.md</code>	Why cascaded multi-task; the leakage story
<code>docs/HPC_HONEST_RERUN_RUNBOOK.md</code>	Reproduce the honest re-run

10. Version history (v1 → v2)

Metric	v1	v2 (honest, clean splits)
Clinical specificity	0% (segmented every image)	classifier abstains on healthy skin
Data categories	1 (DFU only)	3 (DFU + healthy + non-DFU)
Data cleaning / leakage audit	none	full pipeline, <code>has_any_leakage: False</code>
Segmentation Dice	91.7% (inflated, DFU-only, uncleaned)	0.891 DFU-only / 0.853 CV / 0.929 median mixed
Classification accuracy	n/a	0.984
Boundary metrics (HD95, NSD)	not measured	11.3 px / 0.972 (DFU)
Skin-tone fairness	not measured	0.00 gap (DFU-only)
Calibration	not measured	ECE 0.0075 (post temp. scaling)

v1's 91.7% Dice was measured on uncleaned, ulcer-only data with no negative examples. v2's lower numbers are the honest cost of measuring the right thing.